Olympiad Geometry – VII

Inversion

The most exotic geometric transformation we shall cover is inversion.
Unlike the transformations we have seen so far, when applying inversion, figures may substantially change their shape. Yet, as we will see, inversion is an ultimately powerful tool in solving geometric problems.
Properties of inversion can be stated more efficiently if we introduce a point at infinity. We shall denote it as ∞ and we establish that it lies on each line. This extended plane is called the inversion plane.

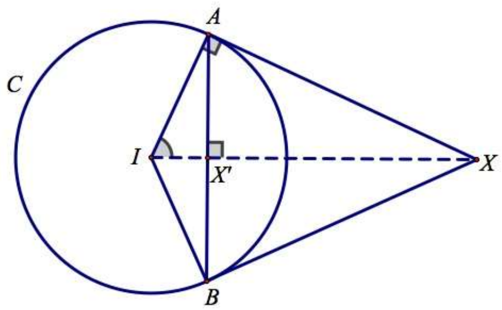
Definition

Take your favourite circle C with centre I and radius $r > 0$.
We define the image X' of point X under inversion about C as follows:

(1) If $X = I$, then $X' = \infty$.
(2) If $X = \infty$, then $X' = I$.
(3) Otherwise, X' is such point on ray IX such that $IX \cdot IX' = r^2$.

Observe that points inside C (with $IX < r$) are mapped to the outside ($IX' > r$) and vice versa, while C is left intact. Further, if we perform inversion about the same circle twice, we obtain identity mapping (nothing happens). In other words, X' is the image of X if and only if X is the image of X'.

Let X be a point outside the circle C centred at I. Let tangents from X touch C at points A, B. Finally, denote by X' the midpoint of AB. Then X' is the image of X under inversion about C.



Theorem

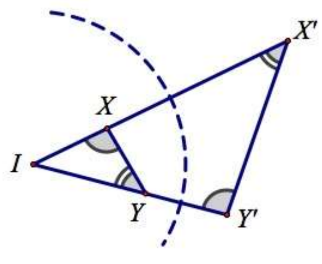
Let I, X, Y be pairwise distinct non-collinear points. Denote by X' and Y' the images of X and Y under inversion about a circle with centre I and radius $r > 0$.

Then $\Delta XIY \sim \Delta Y'IX'$ with ratio of similitude $\dfrac{X'Y'}{XY} = \dfrac{r^2}{IX \cdot IY}$. In particular,

(a) $\angle XYI = \angle IX'Y'$.

(b) $X'Y' = XY \cdot \dfrac{r^2}{IX \cdot IY}$.

(c) $XY = X'Y' \cdot \dfrac{r^2}{IX' \cdot IY'}$.



As we will see, the radius of inversion may often be chosen arbitrarily. In such case, we shall use the notion of inverting about a point. The radius will be considered to be equal to 1.
Now let's see what happens to lines and circles after inversion. The answer is surprisingly convenient!

Image of lines under inversion

Denote by L' the image of line L under inversion about I.

(a) If $I \in L$, then $L' = L$.

(b) If $I \notin L$, then L' is a circle with centre O passing through I such that $OI \perp L$.

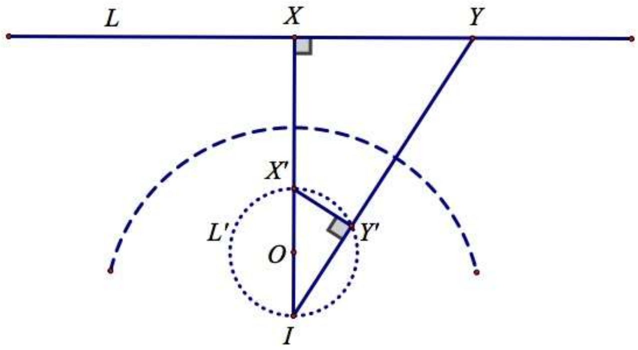
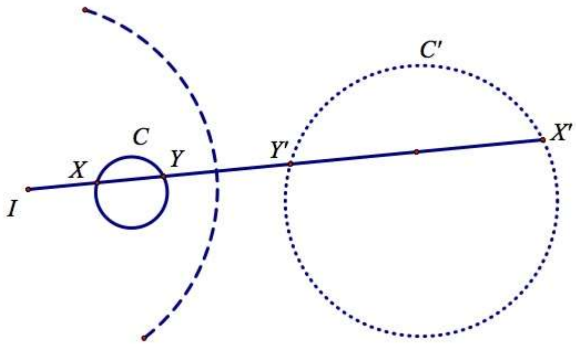


Image of circles under inversion

Denote by C' the image of circle C with centre O under inversion about I.

(a) If $I \in C$, then C' is a line perpendicular to OI.

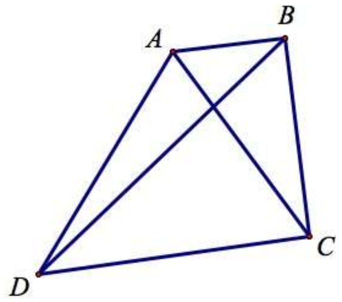
(b) If $I \notin C$, then C' is a circle. Moreover, centres of C and C' are collinear with I.



The idea of using inversion is that we invert both the figure and the desired conclusion to obtain an equivalent problem. Very often (but not always!) this equivalent problem is far easier to solve. As we will see that inverting about a point with many circles passing through it usually leads to much simpler figure.

Example 1 (Ptolemy's Inequality)

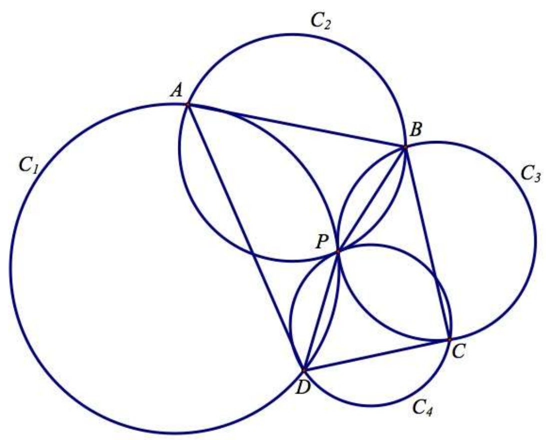
The mutual distances of four distinct points A, B, C, D satisfy $AB \times CD + BC \times AD \geq AC \times BD$ with the equals sign only when A, B, C, D lie on a circle (or on a line) in such an order that either of the arcs AC (or the line segment AC) contains one but not both of the remaining points B and D.



Example 2 (IMO 2003 shortlist)

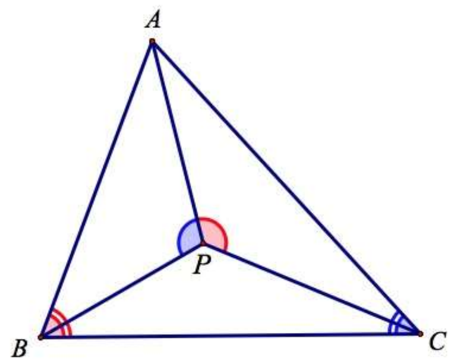
Let C_1, C_2, C_3, C_4 be distinct circles such that C_1, C_3 are externally tangent at P, and C_2, C_4 are externally tangent at the same point P. Suppose that C_1 and C_2, C_2 and C_3, C_3 and C_4, C_4 and C_1 meet at A, B, C, D respectively, and that all these points are different from P. Prove that

$$\frac{AB \cdot BC}{AD \cdot DC} = \frac{PB^2}{PD^2}$$



Example 3 (IMO 1996)

Let P be a point inside a triangle ABC such that $\angle APB - \angle ACB = \angle APC - \angle ABC$.
Let D, E be the incentres of triangles APB, APC, respectively.
Show that the lines AP, BD, CE meet a point.



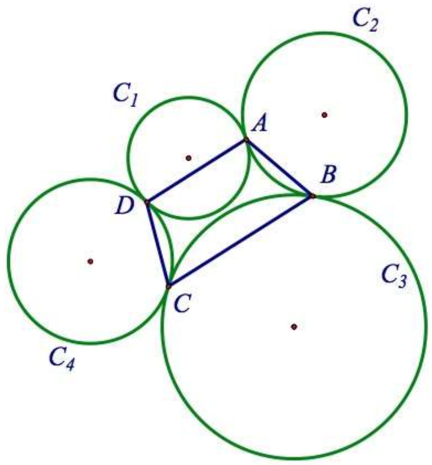
Example 4 (APMO 1994)

Is there an infinite set of points of the plane such that no three elements of it are collinear, and the distance between any two of them is rational?

Problems

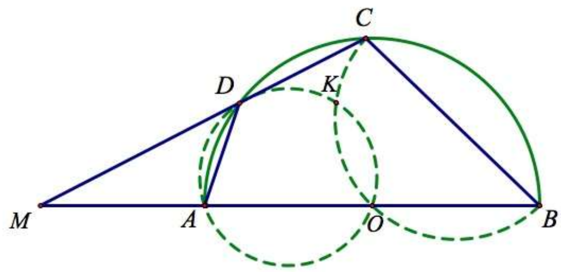
Problem 1

Circles C_1, C_2, C_3 and C_4 are positioned in a such a way that C_1 is tangent to C_2 at point A, C_2 is tangent to C_3 at point B, C_3 is tangent to C_4 at point C, and C_4 is tangent to C_1 at point D. Show that A, B, C and D are either collinear or concyclic.



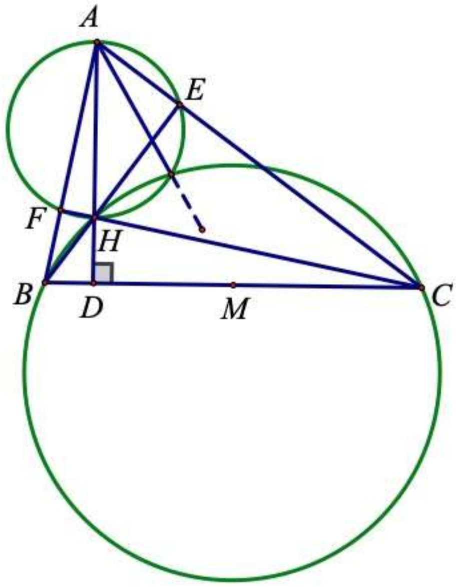
Problem 2 (1995 Russian Math Olympiad)

Let $ABCD$ be a quadrilateral inscribed in a semicircle with diameter AB and centre O. Lines CD and AB intersect at M. Let K be the second point of intersection of the circumcircles of triangles AOD and COB. Prove that $\angle MKO = 90^\circ$.



Problem 3

In ΔABC with orthcentre H, prove that the circle with diameter AH and the circumcircle ΔBHC intersect again on the A-median.

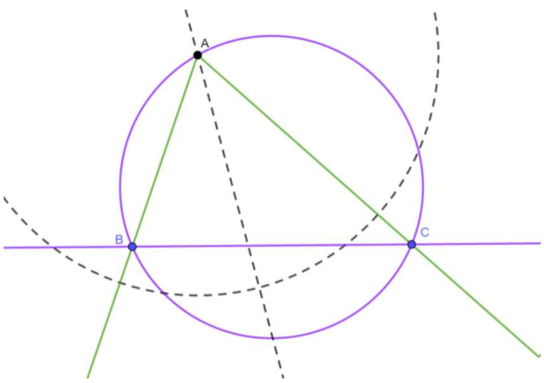


$\sqrt{bc}$ -inversion

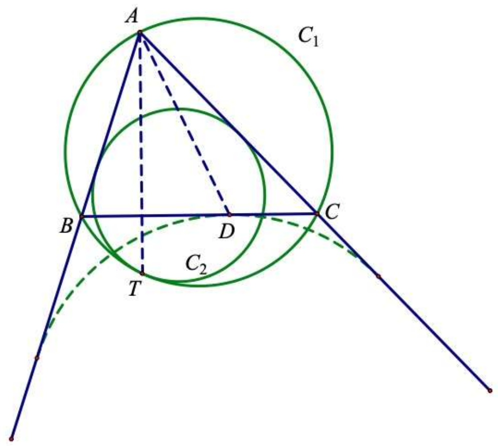
Given a ΔABC, we consider the transformation which first reflects point X over the A-angle bisector, then inverts about A with radius $\sqrt{AB \cdot AC}$ into X' the image of X in $\sqrt{bc}$ -inversion. Complicated at the first glance, but such a trick is "tailor-made" to the triangle such that

1. B maps to C and C maps to B; and
2. the circumcircle of ΔABC maps to line BC, line BC maps to the circumcircle of ΔABC.

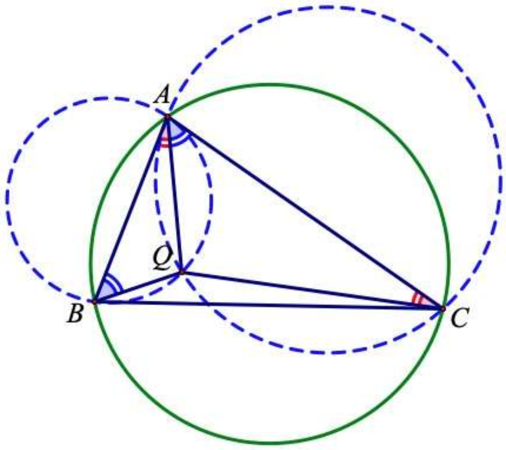
Note that lines AX and AX' are isogonal, so it is comfy to use such a trick in isogonal lines.



Problem 4 (Mixtilinear Incircle)
Let C_1 be the circumcircle of ΔABC. Circle C_2 is tangent to AB and AC, and it touches C_1 internally at T. Let D be the point of tangency of BC and the A-excircle. Show that $\angle BAT = \angle DAC$.



Problem 5 (Dumpty-point)
In ΔABC, let S be a point on the circumcircle of ΔABC such that AS is the A-symmedian. Prove that the midpoint of AS is exactly the A-dumpty point, i.e. the point Q satisfying $\angle BAQ = \angle ACQ$ and $\angle CAQ = \angle ABQ$.



Problem 6

Let ABC be a triangle wher X is the A-excentre and I is the incentre. Let M be the circumcentre of ΔBIC, and let G be the point on BC such that $XG \perp BC$. Construct a circle with diameter AX. If the circle with diameter AX and the circumcircle of ΔABC intersect at a second point P, prove that M, G and P are collinear.